Chapter 7. Initial Orbit Determination

7.1 Historical Background

- Initial Orbit Determination: First time development of orbital elements for a body in motion in space
- We define the overall topic of ***initial orbit determination*** as the first-time development of orbital elements for a body in motion in space. For, this lecture, our main application is the Earth-orbiting satellites; however, historical applications focused entirely on the motion of moons, planets, and asteroids, comets.
- Newton's method (Halley used) depended on graphical method.
- Leonhard Euler (1707-1783) developed the first analytical method for parabolic orbits.
- Johann H. Lambert (1728-1779) generalized Euler's formulas to include elliptical and hyperbolic orbits.
- → solving for the orbit between two known position vectors (today known as Lambert's problem)
- Pierre-Simon de Laplace (1749-1827) published a new method of orbit determination based on only on ***angles***.
- Karl Friedrich Gauss (1777-1855) developed a ***practical*** technique for initial orbit determination. → rediscovery of asteroid Ceres.
- Josiah Gibbs (1839-1903) gave an improvement to Gauss's method.
- Samuel Herrick (1911-1974) developed a derivative application – a Taylor series approach to the Gibbs problem. It allows us to solve for very short time spans, which is necessary for modern radar systems and dense observational data.

7.2 Observations of Range, Azimuth, and Elevation

7.2.1 Application: Site-Track

- A common way of determining the position and velocity vector of a satellite from sensor data

- Data usually consists of:
  - Range, Azimuth and Elevation (ρ, β, el)
  - Rates of Range, Azimuth and Elevation $(\dot{\rho}, \dot{\beta}, \dot{el})$
  - Times of Each Observation

- Site Location: Geodetic Latitude, Local Sidereal Time, Height above Sea Level $(\phi_{gd}, \theta_{LST}, h_{ellp})$

- Given $(\phi_{gd}, \theta_{LST}, h_{ellp}, \rho, \beta, el, \dot{\rho}, \dot{\beta}, \dot{el})$,

  Find $(\vec{r}_{IJK}, \vec{v}_{IJK}, \vec{r}_{siteIJK})$

- Determination of $\vec{r}_{siteIJK}$

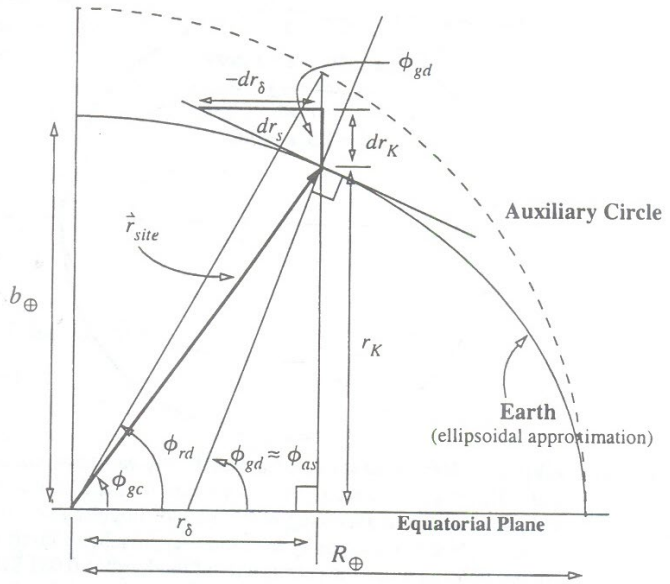


Figure 3-3. **Latitude Geometry (exaggerated scale).** The geodetic latitude, ϕ_{gd}, makes an angle perpendicular to the surface and the equatorial plane, whereas the geocentric latitude, ϕ_{gc}, is referenced to the center of the Earth. Astronomic latitude, ϕ_{as}, is very close to ϕ_{gd}. The center of the ellipsoid differs slightly for ϕ_{gd} and ϕ_{as}. Reduced latitude, ϕ_{rd}, is used only to derive relations.

$$C_\oplus = \frac{R_\oplus}{\sqrt{1 - e_\oplus^2 \sin\phi_{gd}}}, \quad S_\oplus = C_\oplus(1 - e_\oplus^2)$$

$e_\oplus$: Earth's Flattening

$$r_\delta = (C_\oplus + h_{ellp})\cos\phi_{gd}, \quad r_K = (S_\oplus + h_{ellp})\sin\phi_{gd}$$

(Refer to Chapter 3.2.2)

$$\vec{r}_{siteIJK} = \begin{bmatrix} r_\delta \cos\theta_{LST} \\ r_\delta \sin\theta_{LST} \\ r_K \end{bmatrix}$$

$$\theta_{LST} = \theta_{GMST} + \lambda(longitude)$$

(LST: Local Sidereal Time, GMST: Greenwich Mean Sidereal Time)

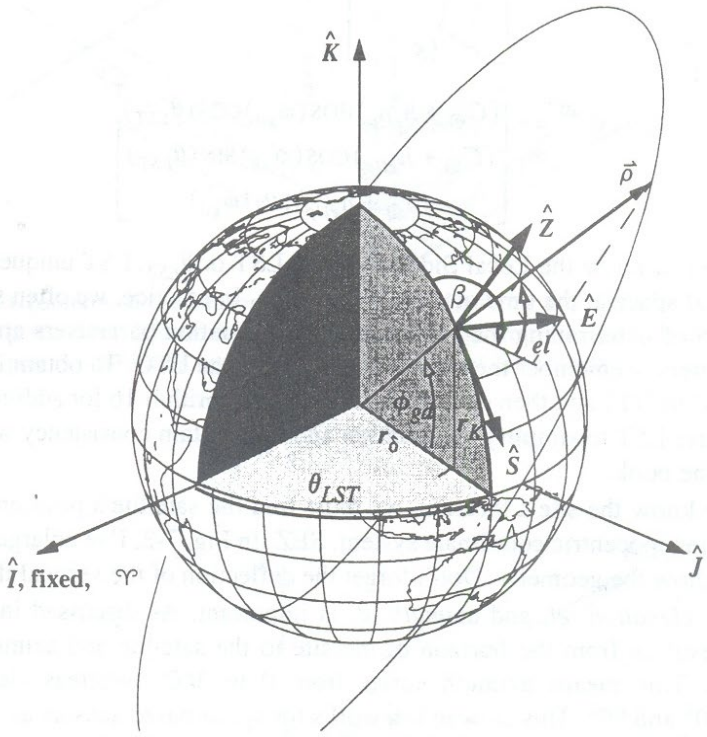
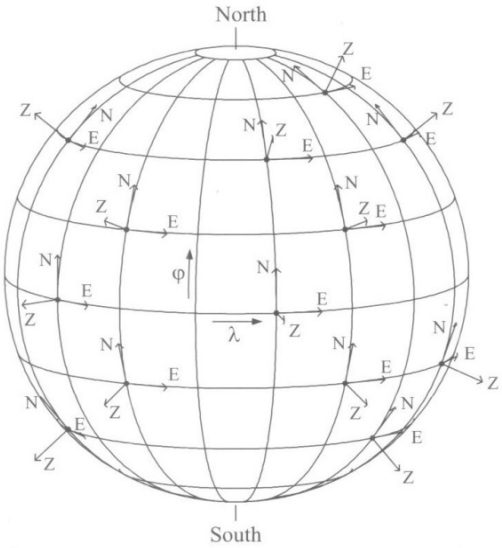


Figure 7-1. **Geometry of Site Coordinates.** The site's position depends on the r_δ and r_K components. Notice we must know LST and site latitude. Geodetic latitude is common, but other conventions exist.

- Determination of $(\vec{r}_{IJK}, \vec{v}_{IJK})$

$$\vec{r}_{IJK} = \vec{\rho}_{IJK} + \vec{r}_{siteIJK}$$
$$\vec{v}_{IJK} = \dot{\vec{\rho}}_{IJK} + \vec{\omega}_{\oplus} \times \vec{r}_{siteIJK} \qquad (\textit{will be proved later})$$

  - $\dot{\vec{\rho}}_{IJK}$: range rate with respect the inertial frame
  - $\vec{\omega}_{\oplus}$: Earth's rotation rate with respect to the inertial frame



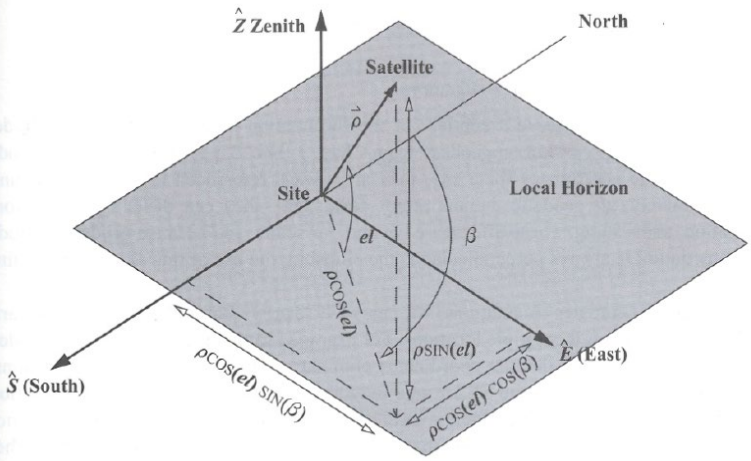


Figure 7-2. **Geometry of the Sensor Site.** This figure shows the site geometry from a different perspective. Elevation is measured positive up from the local horizon. Notice the positive definition because we measure azimuth from north.

$$\vec{\rho}_{SEZ} = \begin{bmatrix} -\rho\cos(el)\cos\beta \\ \rho\cos(el)\sin\beta \\ \rho\sin(el) \end{bmatrix} \text{ in Topocentric Horizon Coordinate (SEZ)}$$

$$\dot{\vec{\rho}}_{SEZ} = \begin{bmatrix} -\dot{\rho}\cos(el)\cos\beta + \rho\sin(el)\cos\beta(\dot{el}) + \rho\cos(el)\sin\beta\dot{\beta} \\ \dot{\rho}\cos(el)\sin\beta - \rho\sin(el)\sin\beta(\dot{el}) + \rho\cos(el)\cos\beta\dot{\beta} \\ \dot{\rho}\sin(el) + \rho\cos(el)(\dot{el}) \end{bmatrix}$$

$$\vec{\rho}_{IJK} = [ROT3(-\theta_{LST})][ROT2(-(90^\circ - \varphi_{gd}))]\vec{\rho}_{SEZ}$$

$$= \begin{bmatrix} cos(-\theta_{LST}) & sin(-\theta_{LST}) & 0 \\ -sin(-\theta_{LST}) & cos(-\theta_{LST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos(-(90^\circ - \varphi_{gd})) & 0 & -sin(-(90^\circ - \varphi_{gd})) \\ 0 & 1 & 0 \\ sin(-(90^\circ - \varphi_{gd})) & 0 & cos(-(90^\circ - \varphi_{gd})) \end{bmatrix} \vec{\rho}_{SEZ}$$

$$= \begin{bmatrix} sin\varphi_{gd}cos\theta_{LST} & -sin\theta_{LST} & cos\varphi_{gd}cos\theta_{LST} \\ sin\varphi_{gd}sin\theta_{LST} & cos\theta_{LST} & cos\varphi_{gd}sin\theta_{LST} \\ -cos\varphi_{gd} & 0 & sin\varphi_{gd} \end{bmatrix} \vec{\rho}_{SEZ}$$

$$\equiv \left[\frac{IJK}{SEZ}\right] \vec{\rho}_{SEZ}$$

$$\dot{\vec{\rho}}_{IJK} = \left[\frac{IJK}{SEZ}\right] \dot{\vec{\rho}}_{SEZ}$$

$$\vec{r}_{IJK} = \vec{\rho}_{IJK} + \vec{r}_{siteIJK}$$

$$\frac{d}{dt}\vec{r}_{IJK} = \frac{d}{dt}\left(\vec{\rho}_{IJK} + \vec{r}_{siteIJK}\right)$$

$$\vec{v}_{IJK} = \dot{\vec{\rho}}_{IJK} + \vec{\omega} \times \vec{r}_{siteIJK}$$

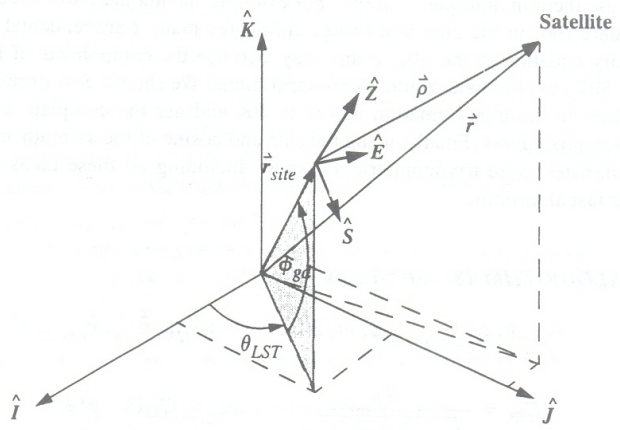


Figure 7-3. **Site-to-Satellite Geometry.** To complete ***SITE-TRACK***, we must use the slant-range vector and the site vector and rotate through geodetic latitude, ϕ_{gd}, and local sidereal time, θ_{LST}.

7.3 Angles-Only Observations

- Optical Sensors essentially take photos of Regions of the Sky
  - Given Angle Data only,

    Find $(\vec{r}, \vec{v})$

  - Fundamentally deficient because of No Range information

- Methodologies
  - Laplace Method:
    - Fits only the middle point but is usually close to the other data points
    - Works poorly for near-Earth satellites
  - Gaussian Method
    - Fits data to all three points and is valid for all the data
    - Better suited for near-Earth satellites, but the formulation limits the spread of data (usually less than $60°$ apart)
  - Double R-iteration
    - Effective for large spreads in the data
    - Good approach for general applications

- Data
  - The azimuth (β) and elevation (el) angles are measured at three different times by an observer through a telescope or other apparatus
  - Azimuth and Elevation angles can be converted into Topocentric Right Ascention (α_t) and Declination (δ_t) by the algorithm 28, p 272, Vallado:

$$\left([\beta, el]_{t=t_1}, [\beta, el]_{t=t_2}, [\beta, el]_{t=t_3}\right) \Rightarrow \left([\alpha_t, \delta_t]_{t=t_1}, [\alpha_t, \delta_t]_{t=t_2}, [\alpha_t, \delta_t]_{t=t_3}\right)$$

  - Topocentric $[\alpha_t, \delta_t]$ values use a plane parallel to the Earth's equator but located at a particular site

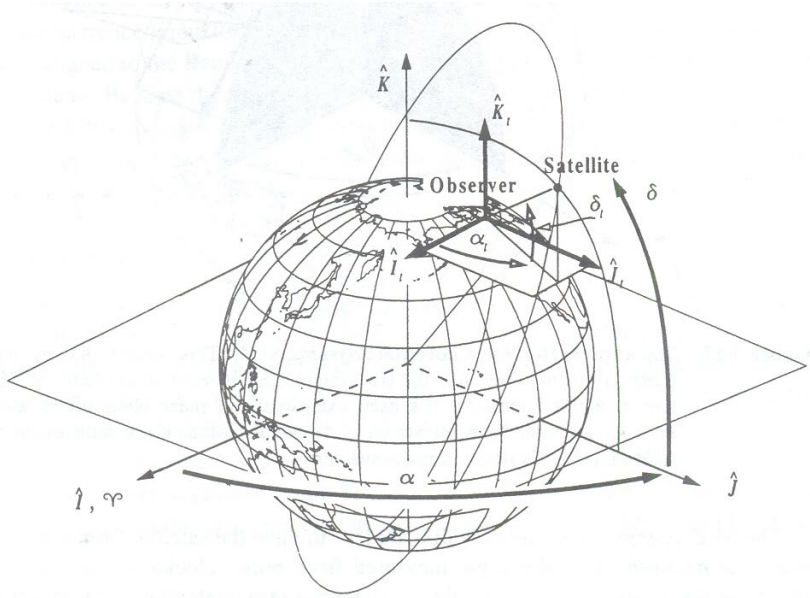


Figure 3-13. Topocentric Equatorial Coordinate System, $I_tJ_tK_t$. This system uses the orientation axes of the IJK system, but it's translated to a particular site. The origin rotates with the Earth but maintains the fixed orientation. The difference in numerical values (δ_t and δ) is apparent in the figure. The topocentric declination, δ_t, in this figure is about 20°, whereas the declination, δ, is about 50°.

- Determination of $(\vec{r}, \vec{v})$

  - Line of Sight unit vector, $\hat{L}$

$$\hat{L}_i = \begin{bmatrix} \cos\delta_{ti}\cos\alpha_{ti} \\ \cos\delta_{ti}\sin\alpha_{ti} \\ \sin\delta_{ti} \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix}, i = 1,2,3 \tag{1}$$

  - If we know the slant range ρ to each satellite and the location of observation site, the position vector is given by

$$\vec{r} = \rho\hat{L} + \vec{r}_{site} \tag{2}$$

(if $\hat{L}$ and $\vec{r}_{site}$ are known, $\vec{r}$ can be calculated)

$$\vec{r} \cdot \vec{r} = (\rho\hat{L} + \vec{r}_{site}) \cdot (\rho\hat{L} + \vec{r}_{site})$$

$$r^2 = \rho^2 + 2\rho\hat{L} \cdot \vec{r}_{site} + r_{site}^2$$

$$r = \sqrt{\rho^2 + 2\rho\hat{L} \cdot \vec{r}_{site} + r_{site}^2} \tag{3}$$

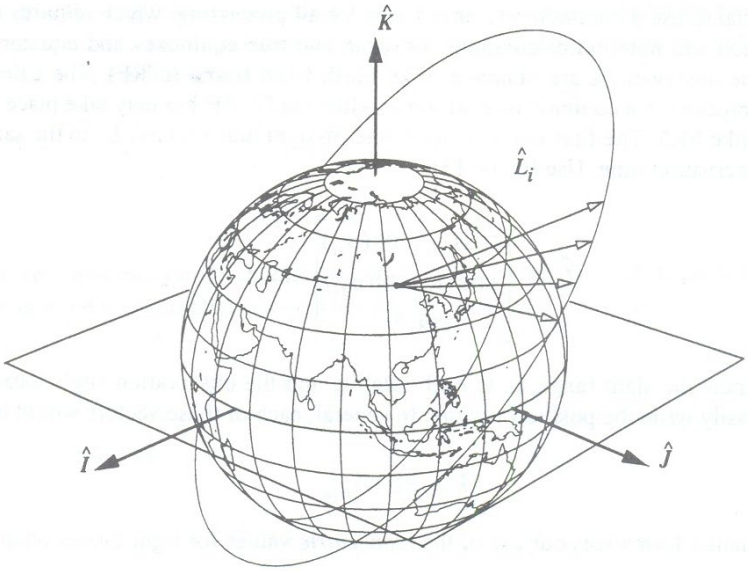


Figure 7-4. Geometry of Angles-only Observations. The underlying principle for the angles-only technique is the use of topocentric angles to form line-of-sight vectors. For objects that are very distant, the distinction for topocentric values diminishes.

7.3.1 Laplace's Method

- Can process data from Different Sites at Different Times (as well as Same Sites at Different Times)
- 3 sets of Angular Measurements = Six Independent Quantities $\Rightarrow$ $(\vec{r}, \dot{\vec{r}} = \vec{v})$

$$\vec{r} = \rho\hat{L} + \vec{r}_{site}$$

$$\dot{\vec{r}} = \dot{\rho}\hat{L} + \rho\dot{\hat{L}} + \dot{\vec{r}}_{site} \tag{4-1}$$

$$\ddot{\vec{r}} = \ddot{\rho}\hat{L} + 2\dot{\rho}\dot{\hat{L}} + \rho\ddot{\hat{L}} + \ddot{\vec{r}}_{site} \tag{4-2}$$

If $\left(\rho, \dot{\rho}, \hat{L}, \dot{\hat{L}}\right)$ are known, then $\left(\vec{r}, \dot{\vec{r}}\right)$ can be calculated

Assume the Two-Body equations of motion:

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} \tag{4-3}$$

Equating (4-2) and (4-3) leads to

$$\ddot{\rho}\hat{L} + 2\dot{\rho}\dot{\hat{L}} + \rho\ddot{\hat{L}} + \ddot{\vec{r}}_{site} = -\frac{\mu}{r^3}(\rho\hat{L} + \vec{r}_{site})$$

$$\Rightarrow \ddot{\rho}\hat{L} + 2\dot{\rho}\dot{\hat{L}} + \rho(\ddot{\hat{L}} + \frac{\mu}{r^3}\hat{L}) = -\frac{\mu}{r^3}\vec{r}_{site} - \ddot{\vec{r}}_{site} \tag{5}$$

Here while $\hat{L}$ is known, all the derivatives in (5) are unknown

An approximate expression for $\hat{L}(t)$ at any time by Interpolation formula:

$$\hat{L}(t) = \frac{(t-t_2)(t-t_3)}{(t_1-t_2)(t_1-t_3)}\hat{L}_1 + \frac{(t-t_1)(t-t_3)}{(t_2-t_1)(t_2-t_3)}\hat{L}_2 + \frac{(t-t_1)(t-t_2)}{(t_3-t_1)(t_3-t_2)}\hat{L}_3$$

$$\left(\Uparrow \quad \vec{A}(t) = \sum_{i=1}^{n} \vec{A}_i \prod_{k \neq i} \frac{(t-t_k)}{(t_i-t_k)} \right), n = 3 \quad \text{for three observations}$$

Differentiating gives

$$\dot{\hat{L}}(t) = \frac{2t - t_2 - t_3}{(t_1 - t_2)(t_1 - t_3)}\hat{L}_1 + \frac{2t - t_1 - t_3}{(t_2 - t_1)(t_2 - t_3)}\hat{L}_2 + \frac{2t - t_1 - t_2}{(t_3 - t_1)(t_3 - t_2)}\hat{L}_3$$

$$\ddot{\hat{L}}(t) = \frac{2}{(t_1-t_2)(t_1-t_3)}\hat{L}_1 + \frac{2}{(t_2-t_1)(t_2-t_3)}\hat{L}_2 + \frac{2}{(t_3-t_1)(t_3-t_2)}\hat{L}_3$$

Simplify by setting the middle time to be zero ($t_2 = 0$) without loss of generality

If all the above observations occur from one site, the derivatives of the site vector are

$$\dot{\vec{r}}_{site2} = \vec{\omega}_{\oplus} \times \vec{r}_{site2}$$
$$\ddot{\vec{r}}_{site2} = \vec{\omega}_{\oplus} \times \dot{\vec{r}}_{site2}$$

Now, we know $\vec{r}_{site2}, \dot{\vec{r}}_{site2}, \ddot{\vec{r}}_{site2}, \hat{L}, \dot{\hat{L}}, \ddot{\hat{L}}$ and (5) becomes for $t_2 = 0$)

$$\begin{bmatrix} \hat{L} & 2\dot{\hat{L}} & \ddot{\hat{L}} + \frac{\mu}{r^3}\hat{L} \end{bmatrix} \begin{bmatrix} \ddot{\rho} \\ \dot{\rho} \\ \rho \end{bmatrix} = -[\ddot{\vec{r}}_{site} + \frac{\mu}{r^3}\vec{r}_{site}] \qquad \text{(3 equations)} \tag{6}$$

Remind you that

$$Ax = b \qquad where \ \ A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Rightarrow \det(A)x_i = \det(A_i), i = 1, \cdots, n \Rightarrow \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} x_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$$

Determinant of the coefficient matrix of (6) is

$$D=\begin{vmatrix} L_x & 2\dot{L}_x & \ddot{L}_x+\frac{\mu}{r^3}L_x \\ L_y & 2\dot{L}_y & \ddot{L}_y+\frac{\mu}{r^3}L_y \\ L_z & 2\dot{L}_z & \ddot{L}_z+\frac{\mu}{r^3}L_z \end{vmatrix}=\begin{vmatrix} L_x & 2\dot{L}_x & \ddot{L}_x \\ L_y & 2\dot{L}_y & \ddot{L}_y \\ L_z & 2\dot{L}_z & \ddot{L}_z \end{vmatrix}=2\begin{vmatrix} L_x & \dot{L}_x & \ddot{L}_x \\ L_y & \dot{L}_y & \ddot{L}_y \\ L_z & \dot{L}_z & \ddot{L}_z \end{vmatrix} \tag{7}$$

Where the first equality is obtained by (third column) – (first column $\times\frac{\mu}{r^3}$) $\Rightarrow$ third column

Apply Cramer's rule:

$$D\rho=2\begin{vmatrix} L_x & \dot{L}_x & -(\ddot{x}_{site}+\frac{\mu}{r^3}x_{site}) \\ L_y & \dot{L}_y & -(\ddot{y}_{site}+\frac{\mu}{r^3}y_{site}) \\ L_z & \dot{L}_z & -(\ddot{z}_{site}+\frac{\mu}{r^3}z_{site}) \end{vmatrix}$$

$$=-2\begin{vmatrix} L_x & \dot{L}_x & \ddot{x}_{site} \\ L_y & \dot{L}_y & \ddot{y}_{site} \\ L_z & \dot{L}_z & \ddot{z}_{site} \end{vmatrix}-2\frac{\mu}{r^3}\begin{vmatrix} L_x & \dot{L}_x & x_{site} \\ L_y & \dot{L}_y & y_{site} \\ L_z & \dot{L}_z & z_{site} \end{vmatrix}$$

$$\equiv -2D_1-2\frac{\mu}{r^3}D_2$$

$$\Rightarrow \rho=-\frac{2D_1}{D}-\frac{2\mu D_2}{r^3D},\ D\neq 0 \tag{8}$$

$$\Rightarrow \rho_2=\frac{-2D_1}{D}-\frac{2\mu D_2}{r_2^3D}\qquad (range) \tag{9}$$

Introducing (9) into (2) results in

$$\vec{r}_2=\rho_2\hat{L}_2+\vec{r}_{site2}$$

Introducing (9) into (3) gives r_2 (at $t=t_2$) as

$$r_2^2=(-\frac{2D_1}{D}-\frac{2\mu D_2}{r_2^3D})^2+2(-\frac{2D_1}{D}-\frac{2\mu D_2}{r_2^3D})\ \hat{L}_2\cdot\vec{r}_{site2}+r_{site2}^2$$

$$\Rightarrow r_2^8+(\frac{4CD_1}{D}-\frac{4D_1^2}{D^2}-r_{site2}^2)r_2^6+\mu(\frac{4CD_2}{D}-\frac{8D_1D_2}{D^2})r_2^3-\frac{4\mu^2D_2}{D^2}=0,\quad C=\hat{L}_2\cdot\vec{r}_{site2}$$

$$\Rightarrow r_2$$

Similar process for the velocity leads to

$$D\dot{\rho} = \begin{vmatrix} L_x & -(\ddot{x}_{site} + \frac{\mu}{r^3} x_{site}) & \ddot{L}_x + \frac{\mu}{r^3} L_x \\ L_y & -(\ddot{y}_{site} + \frac{\mu}{r^3} y_{site}) & \ddot{L}_y + \frac{\mu}{r^3} L_y \\ L_z & -(\ddot{z}_{site} + \frac{\mu}{r^3} z_{site}) & \ddot{L}_z + \frac{\mu}{r^3} L_z \end{vmatrix}$$

$$D\dot{\rho} = -\begin{vmatrix} L_x & \ddot{x}_{site} & \ddot{L}_x \\ L_y & \ddot{y}_{site} & \ddot{L}_y \\ L_z & \ddot{z}_{site} & \ddot{L}_z \end{vmatrix} - \frac{\mu}{r_2^3} \begin{vmatrix} L_x & x_{site} & \ddot{L}_x \\ L_y & y_{site} & \ddot{L}_y \\ L_z & z_{site} & \ddot{L}_z \end{vmatrix}$$

$$\Rightarrow \dot{\rho}_2 = -\frac{D_3}{D} - \frac{\mu D_4}{r_2^3 D}, D \neq 0 \quad \text{(Range Rate)} \tag{10}$$

Introducing (9) and (10) into (4-1) leads to

$$\vec{v}_2 = \dot{\rho}\hat{L}_2 + \rho\dot{\hat{L}}_2 + \dot{\vec{r}}_{site2} \tag{11}$$

Finally,

$$(\vec{r}_2, \vec{v}_2) \xrightarrow[transform]{} \text{ 6 orbital elements}$$

7.3.2 Gauss's Technique

- Begins by assuming that the three position vectors lie in a single plane, i.e.,

$$c_1\vec{r}_1 + c_2\vec{r}_2 + c_3\vec{r}_3 = 0 \tag{1}$$

which implies that any position vector lies in the plane spanned by the other two vectors

$$\begin{aligned} \vec{r}_1 \times (1) &\Rightarrow c_3\left(\vec{r}_1 \times \vec{r}_3\right) = -c_2\left(\vec{r}_1 \times \vec{r}_2\right) \\ \vec{\mathrm{r}}_3 \times (1) &\Rightarrow c_1\left(\vec{\mathrm{r}}_1 \times \vec{\mathrm{r}}_3\right) = -\mathrm{c}_2\left(\vec{\mathrm{r}}_2 \times \vec{\mathrm{r}}_3\right) \end{aligned} \tag{2}$$

The representations by *f* and *g* function (by using the middle position and velocity as basis) are

$$\vec{r}_i = f_i\vec{r}_2 + g_i\vec{v}_2, \; i = 1,3 \tag{3}$$

Let $c_2 = -1$ without loss of generality.

Then, introducing (3) into (2) yields c_1 and c_3 as follows:

$$\begin{aligned} c_1 &= \frac{\vec{r}_2 \times \vec{r}_3}{\vec{r}_1 \times \vec{r}_3} \\ &= \frac{\vec{r}_2 \times (f_3\vec{r}_2 + g_3\vec{v}_2)}{(f_1\vec{r}_2 + g_1\vec{v}_2)\times(f_3\vec{r}_2 + g_3\vec{v}_2)} \\ &= \frac{g_3(\vec{r}_2 \times \vec{v}_2)}{f_1 g_3(\vec{r}_2 \times \vec{v}_2) + f_3 g_1(\vec{v}_2 \times \vec{r}_2)} \\ &= \frac{g_3}{f_1 g_3 - f_3 g_1} \end{aligned} \tag{4}$$

$$\begin{aligned} c_3 &= \frac{\vec{r}_1 \times \vec{r}_2}{\vec{r}_1 \times \vec{r}_3} \\ &= \frac{(f_1\vec{r}_2 + g_1\vec{v}_2)\times \vec{r}_2}{(f_1\vec{r}_2 + g_1\vec{v}_2)\times(f_3\vec{r}_2 + g_3\vec{v}_2)} \\ &= \frac{g_1(\vec{v}_2 \times \vec{r}_2)}{f_1 g_3(\vec{r}_2 \times \vec{v}_2) + g_1 f_3(\vec{v}_2 \times \vec{r}_2)} \\ &= \frac{-g_1}{f_1 g_3 - f_3 g_1} \end{aligned} \tag{5}$$

(2-68) allows us to express *f* and *g* function as series with respect to time τ, with t_2 as the initial epoch:

$$f_i = 1 - \frac{1}{2}u\tau_i^2 + \theta(\tau_i^3)$$

$$g_i = \tau_i - \frac{1}{6}u\tau_i^3 + \theta(\tau_i^4), \quad i = 1,3 \quad where\ \tau_i = t_i - t_2, \quad u = \frac{\mu}{r_2^3}$$

which leads to

$$\begin{aligned} f_1 g_3 - f_3 g_1 &\approx \left(1 - \frac{1}{2}u\tau_1^2\right)\left(\tau_3 - \frac{1}{6}u\tau_3^3\right) - \left(1 - \frac{1}{2}u\tau_3^2\right)\left(\tau_1 - \frac{1}{6}u\tau_1^3\right) \\ &= \tau_3 - \frac{1}{6}u\tau_3^3 - \frac{1}{2}u\tau_1^2\tau_3 + \frac{1}{12}u^2\tau_1^2\tau_3^3 - \tau_1 + \frac{1}{6}u\tau_1^3 + \frac{1}{2}u\tau_3^2\tau_1 - \frac{1}{12}u^2\tau_3^2\tau_1^3 \\ &= (\tau_3 - \tau_1) - \frac{1}{6}u(\tau_3^3 - \tau_1^3) - \frac{1}{2}u(\tau_1^2\tau_3 - \tau_3^2\tau_1) \\ &= (\tau_3 - \tau_1) - \frac{1}{6}u(\tau_3^3 - \tau_1^3 + 3\tau_1^2\tau_3 - 3\tau_3^2\tau_1) \\ &= \tau_3 - \tau_1 - \frac{1}{6}u\left(\tau_3 - \tau_1\right)^3 \\ &= \left(\tau_3 - \tau_1\right)\left[1 - \frac{1}{6}u\left(\tau_3 - \tau_1\right)^2\right] \end{aligned}$$

Introducing the above expression into (4) and (5) yields

$$
\begin{aligned}
c_1 &= \frac{g_3}{f_1 g_3 - f_3 g_1} \\
&\approx \frac{\tau_3 - \frac{1}{6} u \tau_3^3}{\left(\tau_3 - \tau_1\right)\left[1 - \frac{1}{6} u \left(\tau_3 - \tau_1\right)^2\right]} \\
&\left(\Uparrow binomial\ \ series\ \ \left(1+x\right)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \ldots \right) \\
&\approx \frac{\tau_3 \left(1 - \frac{1}{6} u \tau_3^2\right)}{\left(\tau_3 - \tau_1\right)} \left[1 + \frac{1}{6} u \left(\tau_3 - \tau_1\right)^2\right] \\
&= \frac{\tau_3}{\left(\tau_3 - \tau_1\right)} \left[1 + \frac{1}{6} u \left(\tau_3 - \tau_1\right)^2 - \frac{1}{6} u \tau_3^2 - \cdots\right] \\
&= \frac{\tau_3}{\left(\tau_3 - \tau_1\right)} \left[1 + \frac{1}{6} \left[\left(\tau_3 - \tau_1\right)^2 - \tau_3^2\right] u - \cdots\right] \\
&\approx \frac{\tau_3}{\left(\tau_3 - \tau_1\right)} + \frac{\tau_3 \left[\left(\tau_3 - \tau_1\right)^2 - \tau_3^2\right]}{6\left(\tau_3 - \tau_1\right)} u \\
&\Rightarrow c_1 \equiv a_1 + a_{1u} u
\end{aligned}
\tag{6}
$$

$$
\begin{aligned}
c_3 &= \frac{-g_1}{f_1 g_3 - f_3 g_1} \\
&\approx -\frac{\tau_1}{(\tau_3 - \tau_1)} - \frac{\tau_1 \left[\left(\tau_3 - \tau_1\right)^2 - \tau_1^2\right]}{6(\tau_3 - \tau_1)} u \\
&\Rightarrow c_3 = a_3 + a_{3u} u
\end{aligned}
\tag{7}
$$

Introducing $\vec{\mathrm{r}} = \rho \hat{\mathrm{L}} + \vec{\mathrm{r}}_{\mathrm{site}}$ into (1) leads to

$$
\begin{aligned}
& c_1(\rho_1 \hat{L}_1 + \vec{r}_{site,1}) + c_2(\rho_2 \hat{L}_2 + \vec{r}_{site,2}) + c_3(\rho_3 \hat{L}_3 + \vec{r}_{site,3}) = \vec{0} \\
& \Rightarrow c_1 \rho_1 \hat{L}_1 + c_2 \rho_2 \hat{L}_2 + c_3 \rho_3 \hat{L}_3 = -c_1 \vec{r}_{site,1} - c_2 \vec{r}_{site,2} - c_3 \vec{r}_{site,3} \\
& \Rightarrow \begin{bmatrix} \hat{L}_1 & \hat{L}_2 & \hat{L}_3 \end{bmatrix} \begin{bmatrix} c_1 \rho_1 \\ c_2 \rho_2 \\ c_3 \rho_3 \end{bmatrix} = \begin{bmatrix} \vec{r}_{site,1} & \vec{r}_{site,2} & \vec{r}_{site,3} \end{bmatrix} \begin{Bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{Bmatrix} \\
& where\ \ \hat{L}_i = L_{xi} \hat{I} + L_{yi} \hat{J} + L_{zi} \hat{K}, \quad \vec{r}_{site,i} = x_{site,i} \hat{I} + y_{site,i} \hat{J} + z_{site,i} \hat{K}
\end{aligned}
$$

$$\Rightarrow \vec{L}\begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \begin{bmatrix} \vec{r}_{site,1} & \vec{r}_{site,2} & \vec{r}_{site,3} \end{bmatrix}\begin{Bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{Bmatrix}$$

$$where\ \vec{L} \equiv \begin{bmatrix} \hat{L}_1 & \hat{L}_2 & \hat{L}_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \vec{L}^{-1}\begin{bmatrix} \vec{r}_{site,1} & \vec{r}_{site,2} & \vec{r}_{site,3} \end{bmatrix}\begin{Bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{Bmatrix}$$

cf)

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}\begin{pmatrix} a_{22}a_{33} - a_{23}a_{32} & a_{13}a_{32} - a_{12}a_{33} & a_{12}a_{23} - a_{13}a_{22} \\ a_{23}a_{31} - a_{21}a_{33} & a_{11}a_{33} - a_{13}a_{31} & a_{13}a_{21} - a_{11}a_{23} \\ a_{21}a_{32} - a_{22}a_{31} & a_{12}a_{31} - a_{11}a_{32} & a_{11}a_{22} - a_{12}a_{21} \end{pmatrix}$$

$$L^{-1} = \frac{\begin{bmatrix} L_{y2}L_{z3} - L_{y3}L_{z2} & L_{x3}L_{z2} - L_{x2}L_{z3} & L_{x2}L_{y3} - L_{x3}L_{y2} \\ L_{y3}L_{z1} - L_{y1}L_{z3} & L_{x1}L_{z3} - L_{x3}L_{z1} & L_{x3}L_{y1} - L_{x1}L_{y3} \\ L_{y1}L_{z2} - L_{y2}L_{z1} & L_{x2}L_{z1} - L_{x1}L_{z2} & L_{x1}L_{y2} - L_{x2}L_{y1} \end{bmatrix}}{|L|}$$

$$where\ \ c_1 = a_1 + a_{1u}u,\ \ c_2 = -1,\ \ c_3 = a_3 + a_{3u}u,$$

$$\Rightarrow \begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \vec{M}\begin{Bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{Bmatrix}$$

$$where\ \ \vec{M} = \vec{L}^{-1}\begin{bmatrix} \vec{r}_{site,1} & \vec{r}_{site,2} & \vec{r}_{site,3} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}\begin{Bmatrix} -a_1 - a_{1u}u \\ 1 \\ -a_3 - a_{3u}u \end{Bmatrix} \tag{8}$$

Then, with $c_2 = -1$, the slant-range equation for the middle time (t_2) is

$$\begin{aligned} \rho_2 &= \left(M_{21}a_1 - M_{22} + M_{23}a_3\right) + \left(M_{21}a_{1u} + M_{23}a_{3u}\right)u \\ &\equiv d_1 + d_2 u \end{aligned} \tag{9}$$

Substituting $\rho_2 = d_1 + d_2 u = d_1 + d_2\left(\mu / r_2^3\right)$ into $r_2 = \sqrt{\rho_2^{\ 2} + 2\rho_2\hat{L}_2 \cdot \vec{r}_{site,2} + r_{site,2}^2}$

gives the 8th order polynomial in r_2:

$$r_2^8 - (d_1^2 + 2d_1\hat{L}_2 \cdot \vec{r}_{site,2} + r_{site,2}^2)r_2^6 - 2\mu(d_2\hat{L}_2 \cdot \vec{r}_{site,2} + d_1 d_2)r_2^3 - \mu^2 d_2^2 = 0 \qquad (10)$$

- Summary: Gauss's Technique
  - Measure 3 sets of azimuth and elevation angles:

    $[\beta, el]_{t=t_1}, [\beta, el]_{t=t_2}, [\beta, el]_{t=t_3}$

  - Transform into the topocentric right ascension and declination (Algorithm 28, p 272, Vallado):

    $[\alpha_t, \delta_t]_{t=t_1}, [\alpha_t, \delta_t]_{t=t_2}, [\alpha_t, \delta_t]_{t=t_3}$

  - Determine the line of sight vector:

    $\begin{bmatrix} \hat{L}_1 & \hat{L}_2 & \hat{L}_3 \end{bmatrix}$

  - Solve (10) to determine r_2

  - Determine $u = \mu / r_2^3$

  - Determine $c_1 = a_1 + a_{1u}u$ and $c_3 = a_3 + a_{3u}u$ from (6) and (7), and recall that $c_2 = -1$ without loss of generality

  - Solve (8) to determine (ρ_1, ρ_2, ρ_3), which in turn gives $(\vec{r}_1, \vec{r}_2, \vec{r}_3)$ by

    $\vec{r} = \rho\hat{L} + \vec{r}_{site}$

  - Given three position vectors and time, use the Gibbs Method or Herrick-Gibbs Method to determine the velocity vector at the middle point, and thus to determine the orbit:

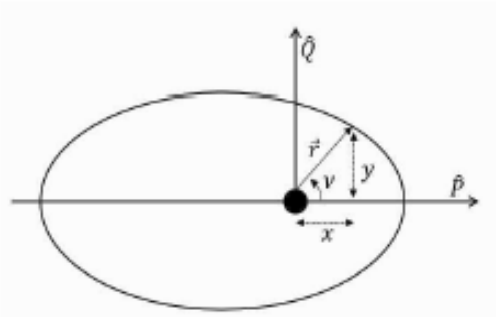
    $\vec{r}_2, \vec{v}_2 \Rightarrow$ 6 orbital elements

  - Terminate for moderate accuracy determination
  - Perform the iteration process in Algorithm 51 for high accuracy determination

Remind you the followings:

Classical Formulas using f and g Functions

Ref: Fundamentals of Astrodynamics and Applications, 3rd ed., David Vallado, 2007, p.90-94



<Geometry for the Perifocal Coordinate System>

● Algorithm

① In the perifocal coordinate system position and velocity vectors can be expressed as

$$\vec{r} = x\hat{P} + y\hat{Q}$$
$$\vec{v} = \dot{x}\hat{P} + \dot{y}\hat{Q}$$

② Also assume $(\vec{r}, \vec{v})$ as a linear combination of $(\vec{r_0}, \vec{v_0})$

$$\vec{r} = f\vec{r_0} + g\vec{v_0}$$
$$\vec{v} = \dot{f}\vec{r_0} + \dot{g}\vec{v_0}$$

③ $\vec{r}\times\vec{v_0} = (f\vec{r_0} + g\vec{v_0})\times\vec{v_0}$
$= f(\vec{r_0}\times\vec{v_0}) + g(\vec{v_0}\times\vec{v_0})$
$= fh\hat{W}$

$$\vec{r}\times\vec{v_0} = \begin{vmatrix} \hat{P} & \hat{Q} & \hat{W} \\ x & y & 0 \\ \dot{x}_0 & \dot{y}_0 & 0 \end{vmatrix} = (x\dot{y}_0 - \dot{x}_0 y)\hat{W}$$

$$\Rightarrow x\dot{y}_0 - \dot{x}_0 y = fh$$

$$\Rightarrow f = \frac{x\dot{y}_0 - \dot{x}_0 y}{h}, \quad \dot{f} = \frac{\dot{x}\dot{y}_0 - \dot{x}_0\dot{y}}{h} \quad \cdots\cdots\cdots\cdots (1)$$

④ $\vec{r_0}\times\vec{r} = \vec{r_0}\times(f\vec{r_0} + g\vec{v_0})$
$= f(\vec{r_0}\times\vec{r_0}) + g(\vec{r_0}\times\vec{v_0})$
$= gh\hat{W}$

$$\vec{r_0}\times\vec{r} = \begin{vmatrix} \hat{P} & \hat{Q} & \hat{W} \\ x_0 & y_0 & 0 \\ x & y & 0 \end{vmatrix} = (x_0 y - x y_0)\hat{W}$$

$$\Rightarrow x_0 y - x y_0 = fh$$

$$\Rightarrow g = \frac{x_0 y - x y_0}{h}, \quad \dot{g} = \frac{x_0\dot{y} - \dot{x}y_0}{h} \quad \cdots\cdots\cdots\cdots (2)$$

(1) and (2) determines $(\vec{r}, \vec{v})$ by

$$\vec{r} = f\vec{r_0} + g\vec{v_0}$$
$$\vec{v} = \dot{f}\vec{r_0} + \dot{g}\vec{v_0}$$

※ Accuracy check

$$\vec{h} = \vec{r}\times\vec{v} = (f\vec{r_0} + g\vec{v_0})\times(\dot{f}\vec{r_0} + \dot{g}\vec{v_0}) = (f\dot{g} - \dot{f}g)\vec{h}$$

$\Leftrightarrow 1 = f\dot{g} - \dot{f}g$ to be checked for accuracy

Assume that,

1) ν is known :

$$x = r\cos\nu \qquad y = r\sin\nu$$
$$\dot{x} = -\sqrt{\frac{\mu}{P}}\sin\nu \qquad \dot{y} = \sqrt{\frac{\mu}{P}}(e + \cos\nu)$$

$$\Rightarrow f = 1 - \frac{r}{P}(1 - \cos\Delta\nu)$$
$$g = \frac{r r_0 \sin\Delta\nu}{\sqrt{\mu P}}$$
$$\dot{f} = \sqrt{\frac{\mu}{P}}\tan\left(\frac{\Delta\nu}{2}\right)\left[\frac{1-\cos\nu}{P} - \frac{1}{r} - \frac{1}{h}\right]$$
$$\dot{g} = 1 - \left(\frac{r_0}{P}\right)(1 - \cos\Delta\nu)$$

2) E is known :

$$x = a[\cos E - e] \qquad y = (a\sin E)\sqrt{1-e^2}$$

① $M = n(t - t_0)$, $\dot{M} = n = \sqrt{\frac{\mu}{a^3}}$

② $M = E - e\sin E$, $\dot{M} = \dot{E}(1 - e\cos E)$

$$\dot{E} = \frac{\dot{M}}{1 - e\cos E} = \sqrt{\frac{\mu}{a^3}}\frac{1}{1 - e\cos E}$$

③ $r = a(1 - e\cos E)$

$$\Rightarrow \dot{E} = \frac{1}{r}\sqrt{\frac{\mu}{a}}$$

$$\Rightarrow \dot{x} = -\frac{\sqrt{\mu a}}{r}\sin E \qquad \dot{y} = \frac{\sqrt{\mu a(1-e^2)}}{r}\cos E$$

$$\Rightarrow f = 1 - \frac{a}{r_0}(1 - \cos\Delta E)$$
$$g = (t - t_0) - \sqrt{\frac{a^3}{\mu}}(\Delta E - \sin\Delta E)$$

$$\dot{f} = \frac{-\sin\Delta E\sqrt{\mu a}}{r_0 r}$$

$$\dot{g} = 1 - \frac{a}{r_0}(1-\cos\Delta E)$$

Series Forms of f and g

Ref: Fundamentals of Astrodynamics and Applications, 3rd ed., David Vallado, 2007, p.94-95

Methods of Orbit Determination, 2nd ed., Pedro R. Escobal, 1975, p.107-111

The series approach permit us to find expressions for f and g, whenever we don't know all the orbital elements.

Especially useful when we know any the position magnitudes(r).

① We attempt to represent the position vector at $t=\tau$ with a Taylor series about $\vec{r}(t=0)$.

$$\begin{aligned} \mathbf{r} = \mathbf{r}_0 &+ \dot{\mathbf{r}}_0\tau + \tfrac{1}{2}\ddot{\mathbf{r}}_0\tau^2 + \tfrac{1}{6}\dddot{\mathbf{r}}_0\tau^3 \\ &+ \tfrac{1}{24}\overset{\mathrm{IV}}{\mathbf{r}}_0\tau^4 + \tfrac{1}{120}\overset{\mathrm{V}}{\mathbf{r}}_0\tau^5 + \tfrac{1}{720}\overset{\mathrm{VI}}{\mathbf{r}}\tau^6 \\ &+ \tfrac{1}{5040}\overset{\mathrm{VII}}{\mathbf{r}}_0\tau^7 + \tfrac{1}{40320}\overset{\mathrm{VIII}}{\mathbf{r}}_0\tau^8 + O(\tau^9). \quad \cdots\cdots (3) \end{aligned}$$

② Need to have attend derivatives at $t=0$

$$\ddot{\mathbf{r}} = -\frac{\mu\mathbf{r}}{r^3}$$

$$\ddot{\mathbf{r}} = -u\mathbf{r}, \qquad u = \frac{\mu}{r^3}$$

$$\ddot{r}_0 = -u_0 r_0$$

$$\begin{aligned} \dddot{\mathbf{r}}_0 &= -\dot{u}_0\mathbf{r}_0 - u_0\dot{\mathbf{r}}_0 \\ \overset{\mathrm{IV}}{\mathbf{r}}_0 &= (-\ddot{u}_0 + u_0^2)\mathbf{r}_0 - 2\dot{u}_0\dot{\mathbf{r}}_0 \\ \overset{\mathrm{V}}{\mathbf{r}}_0 &= (-\dddot{u}_0 + 4u_0\dot{u}_0)\mathbf{r}_0 - (3\ddot{u}_0 - u_0^2)\dot{\mathbf{r}}_0 \\ \overset{\mathrm{VI}}{\mathbf{r}}_0 &= (-\overset{\mathrm{IV}}{u}_0 + 7u_0\ddot{u}_0 - u_0^3 + 4\dot{u}_0^2)\mathbf{r}_0 \\ &\quad - (4\dddot{u}_0 - 6u_0\dot{u}_0)\dot{\mathbf{r}}_0 \\ \overset{\mathrm{VII}}{\mathbf{r}}_0 &= (-\overset{\mathrm{V}}{u}_0 + 15\dot{u}_0\ddot{u}_0 + 11u_0\dddot{u}_0 - 9u_0^2\dot{u}_0)\mathbf{r}_0 \\ &\quad - (5\overset{\mathrm{IV}}{u}_0 - 13u_0\ddot{u}_0 - 10\dot{u}_0^2 + u_0^3)\dot{\mathbf{r}}_0 \\ \overset{\mathrm{VIII}}{\mathbf{r}}_0 &= (-\overset{\mathrm{VI}}{u}_0 + 15\ddot{u}_0^2 + 26\dot{u}_0\dddot{u}_0 + 16u_0\overset{\mathrm{IV}}{u}_0 - 28u_0\dot{u}_0^2 \\ &\quad - 22u_0^2\ddot{u}_0 + u_0^4)\mathbf{r}_0 - (6\overset{\mathrm{V}}{u}_0 - 48\dot{u}_0\ddot{u}_0 \\ &\quad - 24u_0\dddot{u}_0 + 12u_0^2\dot{u}_0)\dot{\mathbf{r}}_0 \end{aligned}$$

$\cdots$ (4)

③ Introducing (4) into (3) and grouping with respect to $(\vec{r}_0, \dot{\vec{r}}_0)$, we have

$$\mathbf{r} = f\mathbf{r}_0 + g\dot{\mathbf{r}}_0$$

where,

$$\begin{aligned} f = 1 &- \tfrac{1}{2}u_0\tau^2 - \tfrac{1}{6}\dot{u}_0\tau^3 \\ &- \tfrac{1}{24}(\ddot{u}_0 - u_0^2)\tau^4 - \tfrac{1}{120}(\dddot{u}_0 - 4u_0\dot{u}_0)\tau^5 \\ &- \tfrac{1}{720}(\overset{\mathrm{IV}}{u}_0 - 7u_0\ddot{u}_0 + u_0^3 - 4\dot{u}_0^2)\tau^6 \\ &- \tfrac{1}{5040}(\overset{\mathrm{V}}{u}_0 - 15\dot{u}_0\ddot{u}_0 - 11u_0\dddot{u}_0 + 9u_0^2\dot{u}_0)\tau^7 \\ &- \tfrac{1}{40320}(\overset{\mathrm{VI}}{u}_0 - 15\ddot{u}_0^2 - 26\dot{u}_0\dddot{u}_0 - 16u_0\overset{\mathrm{IV}}{u}_0 \\ &+ 28u_0\dot{u}_0^2 + 22u_0^2\ddot{u}_0 - u_0^4)\tau^8 \end{aligned}$$

$$\begin{aligned} g = \tau &- \tfrac{1}{6}u_0\tau^3 - \tfrac{1}{12}\dot{u}_0\tau^4 - \tfrac{1}{120}(3\ddot{u}_0 - u_0^2)\tau^5 \\ &- \tfrac{1}{360}(2\dddot{u}_0 - 3u_0\dot{u}_0)\tau^6 \\ &- \tfrac{1}{5040}(5\overset{\mathrm{IV}}{u}_0 - 13u_0\ddot{u}_0 - 10\dot{u}_0^2 + u_0^3)\tau^7 \\ &- \tfrac{1}{40320}(6\overset{\mathrm{V}}{u}_0 - 48\dot{u}_0\ddot{u}_0 - 24u_0\dddot{u}_0 + 12u_0^2\dot{u}_0)\tau^8. \end{aligned}$$

7.3.3 Double *r*-Iteration

7.4 Mixed Observations

7.5 Three Position Vectors and Time

- It happens that:
  - Some observational data, such as range, range-rate, and angle-rate, are missing
  - Only position vectors can be developed
  - ⇒ Need to generate velocity information using only position data to determine orbit.

  - Gibbs Method

    Since the solution is geometrical, it fails when the vectors are closely spaced.
  - Herrick-Gibbs method

    Works best when the vectors are relatively close together, because it is based on a Talyor-series expansion

7.5.1 Gibbs Method $(\vec{r}_1, \vec{r}_2, \vec{r}_3) \Rightarrow \vec{v}_2$

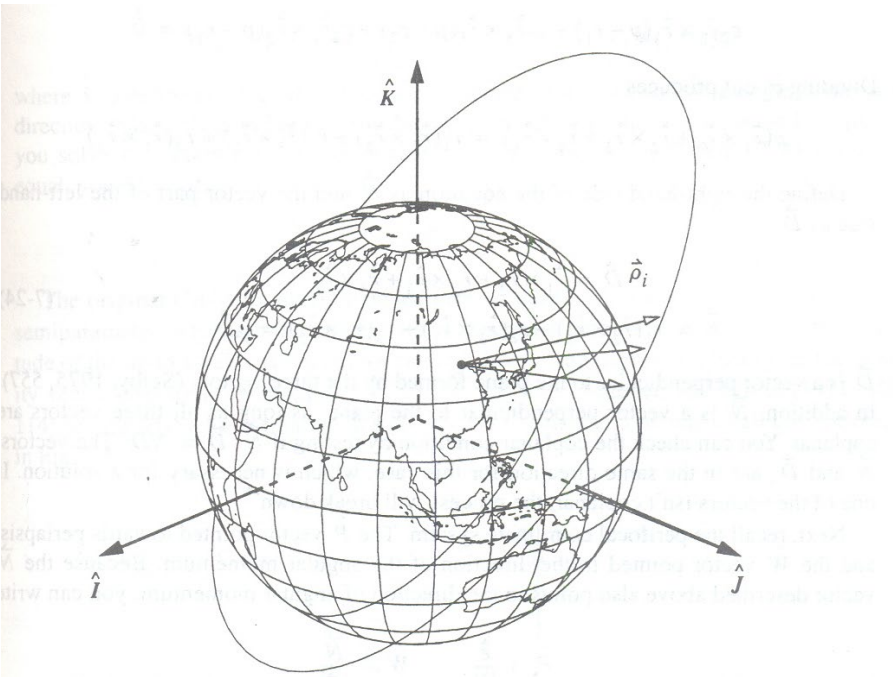


Figure 7-5. Orbit Geometry for the Gibbs Method. To determine the initial orbit using the Gibbs method, we must have at least three sequential observations.

- Begins by assuming that three position vectors are Nonzero, Time-Sequential and Coplanar
  - Step I

$$c_1\vec{r}_1 + c_2\vec{r}_2 + c_3\vec{r}_3 = 0 \qquad (1)$$

$$\vec{r}_1 \times (1) \Rightarrow \quad c_2\vec{r}_1 \times \vec{r}_2 = c_3\vec{r}_3 \times \vec{r}_1$$

$$\vec{r}_2 \times (1) \Rightarrow \quad c_1\vec{r}_1 \times \vec{r}_2 = c_3\vec{r}_2 \times \vec{r}_3 \qquad (2)$$

$$\vec{r}_3 \times (1) \Rightarrow \quad c_1\vec{r}_3 \times \vec{r}_1 = c_2\vec{r}_2 \times \vec{r}_3$$

$$\vec{e} \cdot (1) \quad \Rightarrow c_1\vec{e} \cdot \vec{r}_1 + c_2\vec{e} \cdot \vec{r}_2 + c_3\vec{e} \cdot \vec{r}_3 = 0, \ \vec{e} : \text{eccentricity vector} \qquad (3)$$

$$r = \frac{p}{1 + e\cos\nu}$$

$$\Rightarrow p = r + re\cos\nu$$

$$\Rightarrow p = r + \vec{e} \cdot \vec{r}$$

$$\Rightarrow \vec{e} \cdot \vec{r}_i = p - r_i \qquad (4)$$

Introduce (4) into (3):

$$c_1(p - r_1) + c_2(p - r_2) + c_3(p - r_3) = 0 \qquad (5)$$

Pre-multiply (5) by $(\vec{r}_3 \times \vec{r}_1)$ and introduce (2):

$$c_1(\vec{r}_3 \times \vec{r}_1)(p - r_1) + c_2(\vec{r}_3 \times \vec{r}_1)(p - r_2) + c_3(\vec{r}_3 \times \vec{r}_1)(p - r_3) = 0$$

$$\Rightarrow c_2(\vec{r}_2 \times \vec{r}_3)(p - r_1) + c_2(\vec{r}_3 \times \vec{r}_1)(p - r_2) + c_2(\vec{r}_1 \times \vec{r}_2)(p - r_3) = 0$$

$$\Rightarrow p(\vec{r}_1 \times \vec{r}_2 + \vec{r}_2 \times \vec{r}_3 + \vec{r}_3 \times \vec{r}_1) = r_1(\vec{r}_2 \times \vec{r}_3) + r_2(\vec{r}_3 \times \vec{r}_1) + r_3(\vec{r}_1 \times \vec{r}_2)$$

$$\Rightarrow \ p\vec{D} \equiv \vec{N}$$

where $\vec{D}$ and $\vec{N}$ are vectors perpendicular to the plane formed by the three vectors

$$\Rightarrow \ pD \equiv N \qquad (6)$$

  - Step II

In Perifocal Coordinate System

$\hat{P} = \dfrac{\vec{e}}{e}$ points toward periapsis

$\hat{W} = \dfrac{\vec{N}}{N}$ points in the direction of angular momentum: $\hat{W} \,//\, \vec{N}$

$$\hat{Q} = \hat{W} \times \hat{P} = \frac{\vec{N} \times \vec{e}}{Ne}$$

$$\Rightarrow Ne\hat{Q} = \vec{N} \times \vec{e}$$

$$\Rightarrow Ne\hat{Q} = r_1(\vec{r}_2 \times \vec{r}_3) \times \vec{e} + r_2(\vec{r}_3 \times \vec{r}_1) \times \vec{e} + r_3(\vec{r}_1 \times \vec{r}_2) \times \vec{e}$$

$$\left[\Uparrow\ (\vec{A} \times \vec{B}) \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}\right]$$

$$\Rightarrow Ne\hat{Q} = r_1(\vec{r}_2 \cdot \vec{e})\vec{r}_3 - r_1(\vec{r}_3 \cdot \vec{e})\vec{r}_2 + r_2(\vec{r}_3 \cdot \vec{e})\vec{r}_1 - r_2(\vec{r}_1 \cdot \vec{e})\vec{r}_3 + r_3(\vec{r}_1 \cdot \vec{e})\vec{r}_2 - r_3(\vec{r}_2 \cdot \vec{e})\vec{r}$$

$$\left[\Uparrow\ \vec{r}_i \cdot \vec{e} = p - r_i \ \text{from}(4)\right]$$

$$\Rightarrow Ne\hat{Q} = r_1(p - r_2)\vec{r}_3 - r_1(p - r_3)\vec{r}_2 + r_2(p - r_3)\vec{r}_1 - r_2(p - r_1)\vec{r}_3 + r_3(p - r_1)\vec{r}_2 - r_3(p - r_2)\vec{r}_1$$

$$\Rightarrow Ne\hat{Q} = p\left[(r_2 - r_3)\vec{r}_1 + (r_3 - r_1)\vec{r}_2 + (r_1 - r_2)\vec{r}_3\right]$$

$$\Rightarrow Ne\hat{Q} \equiv p\vec{S}$$

Note that $\hat{Q}$ and $\vec{S}$ are in the same direction:

$$\begin{cases} Ne = pS \\ pD = N \ \ \text{from}(6) \end{cases}$$

$$\Rightarrow pD = \frac{pS}{e}$$

$$\Rightarrow e = \frac{S}{D} \tag{7}$$

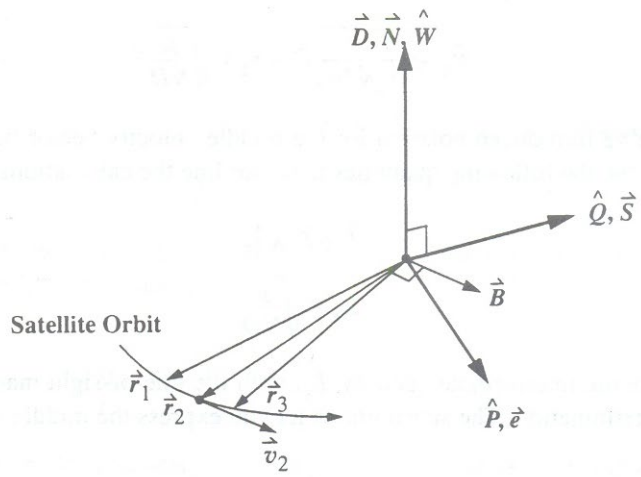


Figure 7-6. Geometry of Vectors for the Gibbs Method. It's useful to visualize the vectors which we use to develop the Gibbs solution. The vectors are not all equal. For instance, P and the eccentricity vector aren't equal, but they're in the same direction. The B vector is normal to $\vec{r}_2$ and $\vec{D}$.

- Step III

  Start from (1-23):

$$\dot{\vec{r}}_2 \times \vec{h} = \mu\left(\frac{\vec{r}_2}{r_2} + \vec{e}\right), \qquad \vec{h}: \text{specific angular momentum}$$

$$\Rightarrow \vec{h}\times(\dot{\vec{r}}_2\times\vec{h}) = \mu\left(\frac{\vec{h}\times\vec{r}_2}{r_2}+\vec{h}\times\vec{e}\right)$$

$$\left[\Uparrow\ (\vec{A}\times\vec{B})\times\vec{C} = (\vec{A}\cdot\vec{C})\vec{B}-(\vec{B}\cdot\vec{C})\vec{A},\ (\vec{A}\times\vec{B})\times\vec{C} = -\vec{C}\times(\vec{A}\times\vec{B})\right]$$

$$\Rightarrow (\vec{h}\cdot\vec{h})\dot{\vec{r}}_2-(\vec{h}\cdot\dot{\vec{r}}_2)\vec{h} = \mu\left(\frac{\vec{h}\times\vec{r}_2}{r_2}+\vec{h}\times\vec{e}\right)$$

$$\left[\Uparrow\ \vec{h}\cdot\dot{\vec{r}}_2 = 0\right]$$

$$\Rightarrow h^2\vec{v}_2 = \mu\left(\frac{\vec{h}\times\vec{r}_2}{r_2}+\vec{h}\times\vec{e}\right)$$

$$\left[\Uparrow\ \vec{h} = h\hat{W},\ \vec{e} = e\hat{P}\right]$$

$$\Rightarrow \vec{v}_2 = \frac{\mu}{h}\left(\frac{\hat{W}\times\vec{r}_2}{r_2}+e\hat{W}\times\hat{P}\right)$$

$$\Rightarrow \vec{v}_2 = \frac{\mu}{h}\left(\frac{\hat{W}\times\vec{r}_2}{r_2}+e\hat{Q}\right)$$

$$\left[\begin{array}{l}\Uparrow\ h = \sqrt{\mu p} = \sqrt{\dfrac{N\mu}{D}}\ \text{ from (6)}\\ e = \dfrac{S}{D}\ \text{ from (7)}\\ \hat{W} = \dfrac{\vec{D}}{D},\quad \hat{Q} = \dfrac{\vec{S}}{S}\end{array}\right]$$

$$\Rightarrow \vec{v}_2 = \frac{\mu}{1}\sqrt{\frac{D}{N\mu}}\left(\frac{\vec{D}\times\vec{r}_2}{Dr_2}+\frac{S}{D}\frac{\vec{S}}{S}\right)$$

$$= \frac{1}{r_2}\sqrt{\frac{\mu}{ND}}\vec{D}\times\vec{r}_2+\sqrt{\frac{\mu}{ND}}\vec{S}$$

$$= \frac{L_g}{r_2}\vec{B}+L_g\vec{S},\ \textit{where}\ \vec{B}\equiv\vec{D}\times\vec{r}_2,\ \ L_g\equiv\sqrt{\frac{\mu}{ND}}$$

- Summary

  Find all three position vectors $(\vec{r}_1,\vec{r}_2,\vec{r}_3)$ by Gauss's technique

  Use Gibbs method to calculate $\vec{v}_2$ using $(\vec{r}_1,\vec{r}_2,\vec{r}_3)$.

  Determine 6 orbital elements for $(\vec{r}_2,\ \vec{v}_2)$

- The angular separation values are of interest because the method rests on geometry. Small angles will cause numerical instability and may yield incorrect answers. The method is robust and works "tolerably" with angles as close as about 1°, but it quickly degrades with smaller angles.

7.5.2 Herrick-Gibbs

- In fact, a typical sensor site's observation of a pass by a satellite usually results in hundreds of observations that are very close together.
  For those occasions where the position vectors are very closely spaced, answers from the Gibbs method are unreliable at best. One solution is the Herrick-Gibbs method which tries to find the middle velocity vector given three sequential position vectors.
- The Herrick-Gibbs method is just a variation of the Gibbs method. The basic premise uses a Taylor series expansion to obtain an expression for the middle velocity vector. Because this method is only approximate, the Herrick-Gibbs method is not as robust as the Gibbs method, and it has a more limited application.

To begin the procedure, expand the position vector as a Taylor series about the middle time t_2.

$$\vec{r}(t) = \vec{r}_2 + \dot{\vec{r}}_2(t-t_2) + \frac{\ddot{\vec{r}}_2(t-t_2)^2}{2!} + \frac{\dddot{\vec{r}}_2(t-t_2)^3}{3!} + \frac{\overset{iv}{\vec{r}}_2(t-t_2)^4}{4!} + \ldots$$

Use this form for the two position vectors, $\vec{r}_1$ and $\vec{r}_3$ with notation for the time differences as $\Delta t_{ij} = t_i - t_j$.

$$\begin{aligned}
\vec{r}_1 &= \vec{r}_2 + \dot{\vec{r}}_2\Delta t_{12} + \frac{\ddot{\vec{r}}_2\Delta t_{12}{}^2}{2!} + \frac{\dddot{\vec{r}}_2\Delta t_{12}{}^3}{3!} + \frac{\overset{iv}{\vec{r}}_2\,\Delta t_{12}{}^4}{4!} + \ldots \\
\vec{r}_3 &= \vec{r}_2 + \dot{\vec{r}}_2\Delta t_{32} + \frac{\ddot{\vec{r}}_2\Delta t_{32}{}^2}{2!} + \frac{\dddot{\vec{r}}_2\Delta t_{32}{}^3}{3!} + \frac{\overset{iv}{\vec{r}}_2\,\Delta t_{32}{}^4}{4!} + \ldots
\end{aligned} \tag{7-28}$$

The first equation $\times(-\Delta t_{32}^2)$ + the second equation $\times\Delta t_{12}^2$ gives

$$\begin{aligned}
-\vec{r}_1\Delta t_{32}{}^2 + \vec{r}_3\Delta t_{12}{}^2 &= \vec{r}_2(-\Delta t_{32}{}^2 + \Delta t_{12}{}^2) + \dot{\vec{r}}_2(-\Delta t_{32}{}^2\Delta t_{12} + \Delta t_{12}{}^2\Delta t_{32}) \\
&+ \frac{\dddot{\vec{r}}_2}{6}(-\Delta t_{32}{}^2\Delta t_{12}^3 + \Delta t_{12}{}^2\Delta t_{32}^3) + \frac{\overset{iv}{\vec{r}}_2}{24}(-\Delta t_{32}{}^2\Delta t_{12}^4 + \Delta t_{12}{}^2\Delta t_{32}^4)
\end{aligned}$$

$$\begin{aligned}
note:\ \Delta t_{12}{}^2\Delta t_{32} - \Delta t_{32}{}^2\Delta t_{12} &= \Delta t_{12}\Delta t_{32}(\Delta t_{12} - \Delta t_{32}) = \Delta t_{12}\Delta t_{32}(t_1 - t_2 - t_3 + t_2) \\
&= \Delta t_{12}\Delta t_{32}\Delta t_{13} \\
\Delta t_{12}{}^2\Delta t_{32}^3 - \Delta t_{32}{}^2\Delta t_{12}^3 &= \Delta t_{12}^2\Delta t_{32}^2\Delta t_{31} \\
\Delta t_{12}{}^2\Delta t_{32}^4 - \Delta t_{32}{}^2\Delta t_{12}^4 &= \Delta t_{12}^2\Delta t_{32}^2\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)
\end{aligned}$$

$$\begin{aligned}
\dot{\vec{r}}_2(\Delta t_{12}\Delta t_{32}\Delta t_{31}) &= \vec{r}_1\Delta t_{32}{}^2 + \vec{r}_2(-\Delta t_{32}{}^2 + \Delta t_{12}{}^2) - \vec{r}_3\Delta t_{12}{}^2 \\
&+ \frac{\dddot{\vec{r}}_2}{6}(\Delta t_{12}{}^2\Delta t_{32}{}^2\Delta t_{31}) + \frac{\overset{iv}{\vec{r}}_2}{24}\left\{\Delta t_{12}{}^2\Delta t_{32}{}^2\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)\right\}
\end{aligned} \tag{7-29}$$

The problem now is to determine expressions for the third and fourth derivatives because everything else in Eq. (7-29) is known. Using Eq. (7-28), differentiate each equation twice, which provides the starting point to remove the higher derivatives.

$$\ddot{\vec{r}}_1 = \ddot{\vec{r}}_2 + \dddot{\vec{r}}_2 \Delta t_{12} + \frac{\overset{iv}{\vec{r}}_2}{2} \Delta t_{12}{}^2 + \ldots$$

$$\ddot{\vec{r}}_3 = \ddot{\vec{r}}_2 + \dddot{\vec{r}}_2 \Delta t_{32} + \frac{\overset{iv}{\vec{r}}_2}{2} \Delta t_{32}{}^2 + \ldots$$

$$-\ddot{\vec{r}}_1 \Delta t_{32} + \ddot{\vec{r}}_3 \Delta t_{12} = \ddot{\vec{r}}_2 \left(-\Delta t_{32} + \Delta t_{12}\right) + \frac{\overset{iv}{\vec{r}}_2}{2} \left(-\Delta t_{32} \Delta t_{12}{}^2 + \Delta t_{12} \Delta t_{32}{}^2\right)$$

$$note: -\Delta t_{32} + \Delta t_{12} = -(t_3 - t_2) + (t_1 - t_2) = -(t_3 - t_1) = -\Delta t_{31} = \Delta t_{13}$$

$$\left(-\Delta t_{32} \Delta t_{12}{}^2 + \Delta t_{12} \Delta t_{32}{}^2\right) = \Delta t_{12} \Delta t_{32} \left(-\Delta t_{12} + \Delta t_{32}\right) = \Delta t_{12} \Delta t_{32} \Delta t_{31}$$

$$-\ddot{\vec{r}}_1 \Delta t_{32} + \ddot{\vec{r}}_3 \Delta t_{12} = \ddot{\vec{r}}_2 \left(-\Delta t_{31}\right) + \frac{\overset{iv}{\vec{r}}_2}{2} \left(\Delta t_{12} \Delta t_{32} \Delta t_{31}\right)$$

$$\overset{iv}{\vec{r}}_2 = \frac{2}{\Delta t_{12} \Delta t_{32} \Delta t_{31}} \left(-\ddot{\vec{r}}_1 \Delta t_{32} + \ddot{\vec{r}}_2 \Delta t_{31} + \ddot{\vec{r}}_3 \Delta t_{12}\right) \qquad (7-30)$$

Similarly, isolate the third derivative by forming

$$-\ddot{\vec{r}}_1 \Delta t_{32}^2 + \ddot{\vec{r}}_3 \Delta t_{12}^2 = \ddot{\vec{r}}_2 \left(-\Delta t_{32}^2 + \Delta t_{12}^2\right) + \dddot{\vec{r}}_2 \left(-\Delta t_{32}^2 \Delta t_{12} + \Delta t_{12}^2 \Delta t_{32}\right) \qquad (7-31)$$

$$\begin{aligned} note: -\Delta t_{32}^2 + \Delta t_{12}^2 &= -(t_3 - t_2)(t_3 - t_2) + (t_1 - t_2)(t_1 - t_2) \\ &= -(t_3 - t_2 + t_1 - t_1)(t_3 - t_2) + (t_1 - t_2 + t_3 - t_3)(t_1 - t_2) \\ &= -\left(t_3 - t_1\right)(t_3 - t_2) - \left(t_1 - t_2\right)(t_3 - t_2) - \left(t_3 - t_1\right)(t_1 - t_2) + \left(t_3 - t_2\right)(t_1 - t_2) \\ &= -\left(t_3 - t_1\right)\left[(t_3 - t_2) + (t_1 - t_2)\right] = -\Delta t_{31} \left(\Delta t_{32} + \Delta t_{12}\right) \end{aligned}$$

$$-\Delta t_{32}^2 \Delta t_{12} + \Delta t_{12}^2 \Delta t_{32} = \Delta t_{12} \Delta t_{32} \left(\Delta t_{12} - \Delta t_{32}\right) = \Delta t_{12} \Delta t_{32} \Delta t_{13}$$

$$-\ddot{\vec{r}}_1 \Delta t_{32}^2 + \ddot{\vec{r}}_3 \Delta t_{12}^2 = \ddot{\vec{r}}_2 \left[-\Delta t_{31} \left(\Delta t_{32} + \Delta t_{12}\right)\right] + \dddot{\vec{r}}_2 \left(\Delta t_{12} \Delta t_{32} \Delta t_{13}\right)$$

$$\dddot{\vec{r}}_2 = \frac{1}{\Delta t_{12} \Delta t_{32} \Delta t_{13}} \left\{-\ddot{\vec{r}}_1 \Delta t_{32}^2 + \ddot{\vec{r}}_2 \left[\Delta t_{31} \left(\Delta t_{32} + \Delta t_{12}\right)\right] + \ddot{\vec{r}}_3 \Delta t_{12}^2\right\} \qquad (7-32)$$

Now, substitute these two expressions for the derivatives into Eq. (7-29), divide through by

$$\dot{\vec{r}}_2 = \frac{1}{\Delta t_{12} \Delta t_{32} \Delta t_{31}} \left\langle \begin{array}{l} \vec{r}_1 \Delta t_{32}{}^2 + \vec{r}_2 (-\Delta t_{32}{}^2 + \Delta t_{12}{}^2) - \vec{r}_3 \Delta t_{12}{}^2 \\ \qquad + \dfrac{\dddot{\vec{r}}_2}{6} (\Delta t_{12}{}^2 \Delta t_{32}{}^2 \Delta t_{31}) + \dfrac{\overset{iv}{\vec{r}}_2}{24} \left\{\Delta t_{12}{}^2 \Delta t_{32}{}^2 \Delta t_{31} \left(\Delta t_{32} + \Delta t_{12}\right)\right\} \end{array} \right\rangle$$

$$
\begin{aligned}
\dot{\vec{r}}_2 &= \frac{1}{\Delta t_{12}\Delta t_{32}\Delta t_{31}}\left[\vec{r}_1\Delta t_{32}{}^2 + \vec{r}_2(-\Delta t_{32}{}^2 + \Delta t_{12}{}^2) - \vec{r}_3\Delta t_{12}{}^2\right] \\
&\quad - \frac{1}{6\Delta t_{31}}\left[\left\{-\ddot{\vec{r}}_1\Delta t_{32}^2 + \ddot{\vec{r}}_2\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right) + \ddot{\vec{r}}_3\Delta t_{12}^2\right\}\right] \\
&\quad + \frac{\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\left(-\ddot{\vec{r}}_1\Delta t_{32} + \ddot{\vec{r}}_2\Delta t_{31} + \ddot{\vec{r}}_3\Delta t_{12}\right) \\
&= \frac{\vec{r}_1\Delta t_{32}}{\Delta t_{12}\Delta t_{31}} - \frac{\vec{r}_2\left(\Delta t_{32} + \Delta t_{12}\right)}{\Delta t_{12}\Delta t_{32}} - \frac{\vec{r}_3\Delta t_{12}}{\Delta t_{32}\Delta t_{31}} + \ddot{\vec{r}}_1\left(\frac{\Delta t_{32}^2}{6\Delta t_{31}} - \frac{\Delta t_{32}\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\right) \\
&\quad + \ddot{\vec{r}}_2\left(\frac{-\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)}{6\Delta t_{31}} + \frac{\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\right) + \ddot{\vec{r}}_3\left(\frac{-\Delta t_{12}^2}{6\Delta t_{31}} + \frac{\Delta t_{12}\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\right) \\
&= \frac{\vec{r}_1\Delta t_{32}}{\Delta t_{12}\Delta t_{31}} - \frac{\vec{r}_2\left(\Delta t_{32} + \Delta t_{12}\right)}{\Delta t_{12}\Delta t_{32}} - \frac{\vec{r}_3\Delta t_{12}}{\Delta t_{32}\Delta t_{31}} + \ddot{\vec{r}}_1\left(\frac{\Delta t_{32}\left(\Delta t_{32} - \Delta t_{12}\right)}{12\Delta t_{31}}\right) \\
&\quad + \ddot{\vec{r}}_2\left(\frac{-\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\right) + \ddot{\vec{r}}_3\left(\Delta t_{12}\frac{-2\Delta t_{12} + \Delta t_{32} + \Delta t_{12}}{12\Delta t_{31}}\right)
\end{aligned}
$$

Eliminate the acceleration vectors by recognizing that we used two-body motion

$$
\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r}
$$

Then we obtain

$$
\begin{aligned}
\dot{\vec{r}}_2 &= \frac{\vec{r}_1\Delta t_{32}}{\Delta t_{12}\Delta t_{31}} - \frac{\vec{r}_2\left(\Delta t_{32} + \Delta t_{12}\right)}{\Delta t_{12}\Delta t_{32}} - \frac{\vec{r}_3\Delta t_{12}}{\Delta t_{32}\Delta t_{31}} + \left(-\frac{\mu}{r_1^3}\vec{r}_1\right)\left(\frac{\Delta t_{32}\left(\Delta t_{32} - \Delta t_{12}\right)}{12\Delta t_{31}}\right) \\
&\quad + \left(-\frac{\mu}{r_2^3}\vec{r}_2\right)\left(\frac{-\Delta t_{31}\left(\Delta t_{32} + \Delta t_{12}\right)}{12\Delta t_{31}}\right) + \left(-\frac{\mu}{r_3^3}\vec{r}_3\right)\left(\Delta t_{12}\frac{-2\Delta t_{12} + \Delta t_{32} + \Delta t_{12}}{12\Delta t_{31}}\right)
\end{aligned}
$$

- Summary
  - Find all three position vectors $(\vec{r}_1, \vec{r}_2, \vec{r}_3)$ by Gauss's technique
  - Use Herrick-Gibbs method to calculate $\vec{v}_2$ using $(\vec{r}_1, \vec{r}_2, \vec{r}_3)$.
  - Determine 6 orbital elements for $\left(\vec{r}_2, \vec{v}_2\right)$